

Neural Internal Model Control: Learning a Robust Control Policy via Predictive Error Feedback

Feng Gao¹, Chao Yu^{13†}, Yu Wang¹, Yi Wu^{12†}

Abstract—Accurate motion control in the face of disturbances within complex environments remains a major challenge in robotics. Classical model-based approaches often struggle with nonlinearities and unstructured disturbances, while reinforcement learning (RL)-based methods can be fragile when encountering unseen scenarios. In this paper, we propose a novel framework, Neural Internal Model Control (NeuralIMC), which integrates model-based control with RL-based control to enhance robustness. Our framework streamlines the predictive model by applying Newton-Euler equations for rigid-body dynamics, eliminating the need to capture complex high-dimensional nonlinearities. This internal model combines model-free RL algorithms with predictive error feedback. Such a design enables a closed-loop control structure to enhance the robustness and generalizability of the control system. We demonstrate the effectiveness of our framework on both quadrotors and quadrupedal robots, achieving superior performance compared to state-of-the-art methods. Furthermore, real-world deployment on a quadrotor with rope-suspended payloads highlights the framework’s robustness in sim-to-real transfer. Our code is released at <https://github.com/thu-uav/NeuralIMC>.

Index Terms—Robust Control, Reinforcement Learning, Sensorimotor Learning

I. INTRODUCTION

ACHIEVING stable and precise motion in complex, disturbed environments is essential for robots, such as quadrotors maintaining stable flight in wind fields and legged robots carrying loads over diverse terrains. This challenge, known as the robust control or adaptive control problem, has been extensively studied [1]. Classical adaptive control methods adjust control parameters in real-time to compensate for dynamic environmental changes, often using mathematical models of the system and disturbances [2], [3], [4], [5], while robust control methods utilize more informative signals to enhance robustness, like Internal Model Control [6]. Despite their guaranteed stability, these methods rely on accurately modeling system dynamics, which is challenging for nonlinear systems with unstructured disturbances. In contrast, reinforcement learning (RL) methods have shown remarkable success in continuous control across various robot morphologies, including skill learning for armed robots [7], [8], high-speed quadrotor racing [9], [10], and adaptive locomotion for

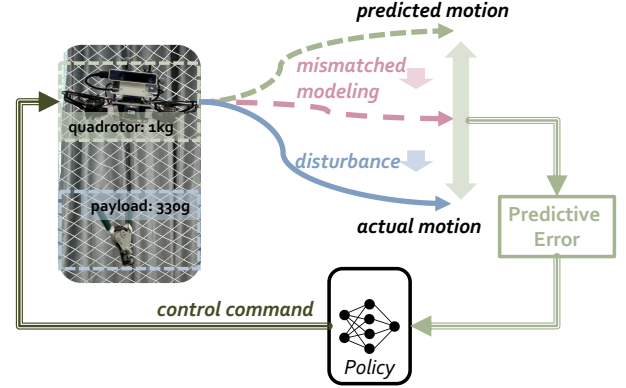


Fig. 1: Predictive errors can be caused by mismatched system modeling and/or external disturbances.

legged robots [11], [12], [13], [14], [15]. Large-scale training with randomized dynamics and environmental parameters enables RL-based control methods to generalize across different conditions and environments [16], [17], allowing robots to perform reliably even in previously unseen and highly varying scenarios.

There has been a recent trend towards integrating model-based control with RL-based neural control to harness the strengths of both worlds [18], [19], [13], [20]. One representative framework is to use predicted future states or estimated disturbances as an additional input signal to obtain a more informative model-free RL policy. Long et al. [13] employ unsupervised contrastive representation learning to train a predictive encoder capable of estimating the next state in the latent space. Conversely, Huang et al. [20] use a handcrafted $\mathcal{L}_1$ adaptive law [21] to explicitly estimate applied force disturbances, resulting in a robust trajectory tracking controller that can handle unexpected wind fields. Although promising, integrating model-based and RL-based controls faces significant challenges. Predictive models often struggle with variability when training data cannot cover real-world dynamics, and handcrafted disturbance estimators require precise tuning, limiting their feasibility for complex systems and adaptability to dynamic changes. For establishing a more general method, we need to develop a structure that effectively combines these controls to fully leverage their advantages and address these limitations.

In this paper, we propose a new learning-based control methods which leverages a predictive error feedback to enhance control robustness, inspired by model-based control. By simplifying the model to focus on the robot’s main body and

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神经内部模型控制：通过预测误差反馈学习鲁棒控制策略

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Abstract——在复杂环境中面对干扰时实现精确的运动控制仍然是机器人技术中的一大挑战。传统基于模型的方法通常在处理非线性和非结构化扰动时表现不佳，而基于强化学习（RL）的方法在遇到未见过的场景时可能会显得脆弱。本文提出了一种新颖的框架——神经内部模型控制（NeuralIMC），将基于模型的控制与基于RL的控制结合，以增强稳健性。我们的框架通过应用刚体动力学的牛顿-欧拉方程简化了预测模型，从而无需捕捉复杂的高维非线性。这一内部模型将无模型RL算法与预测误差反馈结合起来。这种设计使闭环控制结构能够增强控制系统的稳健性和泛化能力。我们在四旋翼和四足机器人上验证了该框架的有效性，实现了优越的性能

Index Terms——鲁棒控制、强化学习、感觉运动学习

一、引言

在复杂且受干扰的环境中，实现机器人（例如在风场中保持稳定飞行的四旋翼和在各种地形上搬运负载的步行机器人）的稳定且精确的运动是至关重要的。这个挑战被称为鲁棒控制或自适应控制问题，已经被广泛研究[1]。经典自适应控制方法通过实时调整控制参数来补偿动态环境变化，通常使用系统和扰动的数学模型[2]、[3]、[4]、[5]，而鲁棒控制方法则利用更多的信息信号来增强鲁棒性，例如内部模型控制[6]。尽管这些方法能够保证稳定性，但它们依赖于对系统动力学的精确建模，而对于具有非线性系统和非结构化干扰的情况，这种建模是具有挑战性的。相比之下，强化学习（RL）方法在各种机器人形态的连续控制中展示了显著的成功，包括武装机器人技能学习[7]。

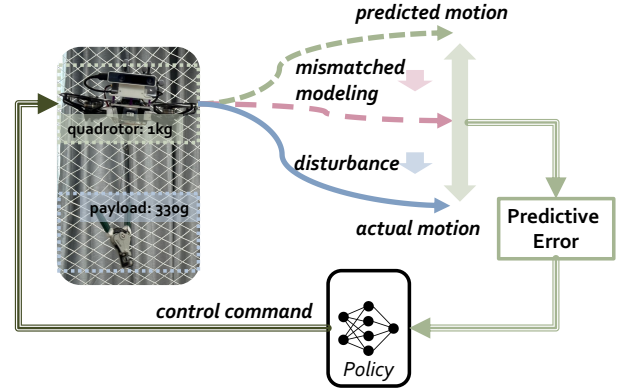


图1：预测误差可能由系统建模不匹配和/或外部扰动引起。

腿足机器人 [11], [12], [13], [14], [15]。通过对动力学和环境参数进行随机化的大规模训练，使基于强化学习的控制方法能够在不同条件和环境下进行泛化 [16], [17]，从而使机器人即使在以前未见过和高度变化的场景中也能可靠地执行任务。

近年来，出现了将基于模型的控制与基于强化学习（RL）的神经网络控制相结合的趋势，以利用两者的优势 [18], [19], [13], [20]。一个典型的框架是使用预测的未来状态或估计的扰动作为附加输入信号，以获得更具信息量的无模型RL策略。Long 等人 [13] 采用无监督对比表示学习来训练能够在潜在空间中估计下一状态的预测编码器。相反，Huang 等人 [20] 使用手工设计的 $\mathcal{L}_1$ 自适应律 [21] 来显式估计施加的力扰动，从而得到一个能够应对意外风场的鲁棒轨迹跟踪控制器。尽管前景可观，将基于模型的控制与基于RL的控制结合仍面临重大挑战。当训练数据无法覆盖现实世界动态时，预测模型通常在变异性方面存在困难，而手工设计的扰动估计器则需要精确调试，从而限制了其可行性。

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在本文中，我们提出了一种新的基于学习的控制方法，该方法利用预测误差反馈来增强控制的鲁棒性，其灵感来源于基于模型的控制。通过简化模型以集中于机器人的主体，

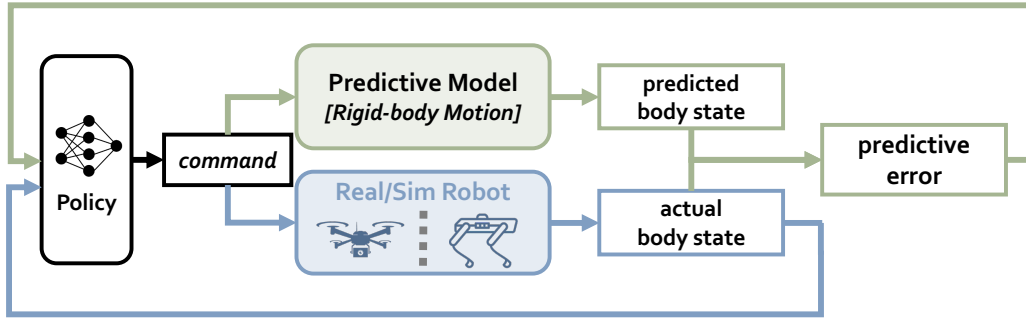


Fig. 2: The closed-loop NeuralIMC framework using an explicit predictive model to calculate the next body state, applicable to both quadrotors and legged robots.

treating its dynamics as a rigid transformation, we bypass the high-dimensional nonlinearity of complex robot systems and environmental interactions. This simplification enables us to use an explicit predictive model without requiring parameter tuning, establishing a foundation for robust closed-loop control. We compute the predictive error—defined as the discrepancy between plant and model responses—as a feedback signal to the policy, enhancing robustness and generalizability through informative feedback and extensive training. Our framework, shown in Fig. 2, can be easily integrated with other learning algorithms by incorporating this feedback loop. To demonstrate the effectiveness and versatility of our method, we apply it to two prevalent robot morphologies: quadrotors, which demand precise 3D control, and quadrupedal robots, which require adaptive locomotion to navigate complex terrain interactions. Empirical results indicate that our framework can surpass recent state-of-the-art control methods in both domains. Furthermore, we conduct a real-world deployment on a quadrotor with various rope-suspended payloads to illustrate its robustness in sim-to-real transfer.

II. RELATED WORK

A. Classical control and disturbance estimation

Classical adaptive controllers require estimating system parameters in a closed-loop manner to adjust control actions in real-time. $\mathcal{L}_1$ adaptive control has been widely used to estimate force disturbances in quadrotors [22], [23]. Neural-Fly [18] incorporated pre-trained representations to enhance rapid online adaptation through deep learning, achieving stable flight in previously unseen wind conditions. Internal Model Control (IMC) [6], [3] is a general structure for robust control that uses a predefined model to predict system responses and estimate disturbances. However, applying IMC to more complex and nonlinear systems is challenging due to the difficulty of crafting reliable models for such systems. Hybrid Internal Model (HIM) further leverages a learning-based internal model that employs unsupervised contrastive learning, with the learned model acting as a predictive encoder for estimating system responses in the latent space, as proposed by Long et al. [13]. Despite their advantages, these methods either rely on explicit disturbance estimation or are heavily influenced by prediction accuracy, which can significantly affect performance if compromised.

B. Learning-based adaptive control

With the rapid advancement of highly parallel simulators [17], [24], [25], reinforcement learning (RL) methods are increasingly demonstrating powerful capabilities to achieve robust control, including quadrotors [26], [20] and legged robots [11], [12], [13], [14], [15]. Unlike classical model-based control, these learning-based methods often use model-free RL algorithms like PPO [27] to train neural controllers in large-scale randomized environments. Koryakovskii et al. [28] proposed to incorporate an RL-based policy to generate residual actions to compensate nonlinear model predictive control in model-mismatch cases. Rapid Motor Adaptation (RMA) [12] is a pure RL-based method, which predicts environmental parameters in the latent space using state-action histories through a teacher-student training procedure. This approach has been extended to quadrotors [26], bipedal robots [29], and humanoid robots [15]. A follow-up study found that a dual-history architecture, leveraging both short-term and long-term input/output history, achieves superior performance with a straightforward one-stage training process [14]. Conversely, DATT [20] incorporates an $\mathcal{L}_1$ adaptation law [21] to estimate forces applied to quadrotors directly, which are then used as policy inputs. CAL [28] integrates RL with nonlinear model predictive control (NMPC) by employing RL to generate residual actions that augment NMPC's outputs. Although both DATT and CAL have proven effective with precise parameter tuning, they remain sensitive to handcrafted settings in their classical control components, limiting their generalizability to more complex systems.

III. METHODS

In this section, we present a general framework that bridges model-based and RL-based control through a simplified predictive model, named NeuralIMC. We start by introducing the basics of Internal Model Control (IMC) and our extension to integrate it with RL-based neural control (Sec. III-A). By leveraging the model-free RL algorithm PPO [27], we reduce the complexity and accuracy requirements typically associated with classical model-based control. Next, we detail how rigid transformation is used to predict the robot's body state, establishing a general approach to estimate disturbances and incorporate them into the control policy (Sec. III-B). Finally, we explain the application of this framework to

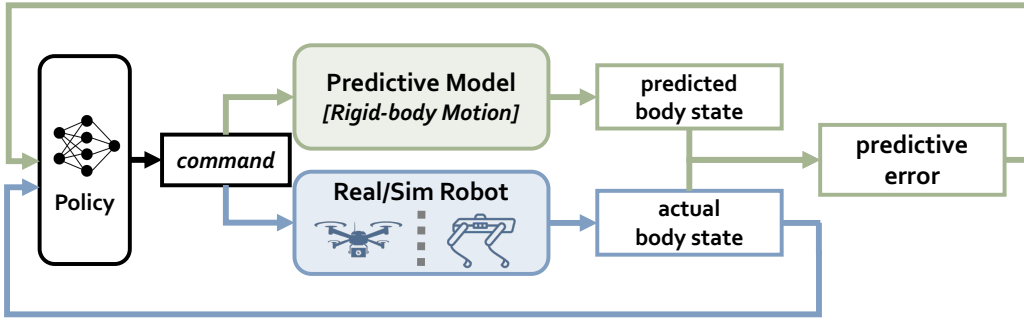


图 2: 使用显式预测模型计算下一个身体状态的闭环 NeuralIMC 框架, applirope 去做四旋翼飞行器和多足机器人。

将其动力学视为刚性变换, 我们绕过了复杂机器人系统和环境交互的高维非线性。这种简化使我们能够使用显式预测模型而无需参数调优, 为稳健的闭环控制建立了基础。我们计算预测误差——定义为系统与模型响应之间的差异——作为策略的反馈信号, 通过提供信息丰富的反馈和广泛的训练来增强稳健性和泛化能力。我们的框架如图2所示, 可以通过引入该反馈回路轻松与其他学习算法集成。为了展示我们方法的有效性和多功能性, 我们将其应用于两种常见的机器人形态: 需要精确三维控制的四旋翼飞行器, 以及需要自适应运动以应对复杂地形交互的四足机器人。实证结果表明, 我们的框架能够超越近期的最先进水平——

B. Learning-based adaptive control

随着高度并行模拟器的快速发展 [17], [24], [25], 强化学习 (RL) 方法在实现稳健控制方面显示出日益强大的能力, 包括四旋翼 [26], [20] 和腿式机器人 [11], [12], [13], [14], [15]。与经典的基于模型的控制不同, 这些基于学习的方法通常使用无模型的 RL 算法, 如 PPO [27], 在大规模随机化环境中训练神经控制器。Koryakovskii 等人 [28] 提出结合基于 RL 的策略生成残差动作, 以在模型不匹配的情况下补偿非线性模型预测控制。快速运动适应 (RMA) [12] 是一种纯粹基于 RL 的方法, 它通过师生训练过程利用状态-动作历史在潜在空间中预测环境参数。这种方法已扩展到四旋翼 [26]、双足机器人 [29] 和人形机器人 [15]。后续研究发现, 利用双历史架构, 同时

二、相关工作

A. Classical control and disturbance estimation

经典自适应控制器需要通过闭环方式估计系统参数, 以实时调整控制动作。 $\mathcal{L}_1$ 自适应控制已被广泛用于估计四旋翼中的力扰动 [22], [23]。Neural-Fly [18] 融入了预训练表示, 通过深度学习增强快速在线适应能力, 实现了在先前未见过的风况下的稳定飞行。内部模型控制 (IMC) [6], [3] 是一种通用的鲁棒控制结构, 它使用预定义模型来预测系统响应和估计扰动。然而, 由于为复杂非线性系统构建可靠模型的难度, 将 IMC 应用于更复杂和非线性系统具有挑战性。混合内部模型 (HIM) 进一步利用基于学习的内部模型, 该模型采用无监督对比学习, 由 Long 等人 [13] 提出, 学习到的模型在潜在空间中充当预测编码器, 用于估计系统响应。尽管它们的 a

三、方法

在本节中, 我们提出了一个通过简化预测模型 (称为 NeuralIMC) 桥接基于模型控制和基于强化学习控制的一般框架。我们首先介绍内部模型控制 (IMC) 的基础知识, 以及我们将其与基于强化学习的神经控制相结合的扩展 (第 III-A 节)。通过利用无模型强化学习算法 PPO [27], 我们降低了经典基于模型控制通常所需的复杂性和精度要求。接下来, 我们详细说明如何使用刚性变换来预测机器人的身体状态, 从而建立了一种估计扰动并将其纳入控制策略的一般方法 (第 III-B 节)。最后, 我们解释了该框架的应用。

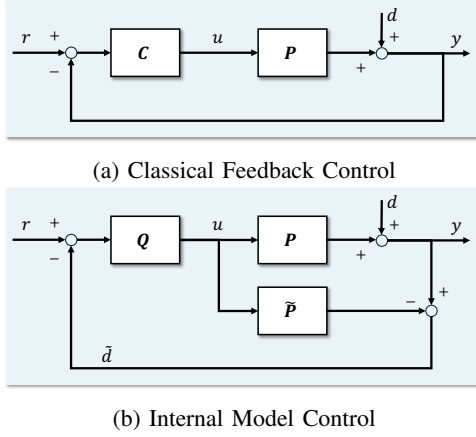


Fig. 3: Feedback structures of classical feedback control and Internal Model Control (IMC).

two specific robot morphologies: quadrotors and quadrupedal robots (Sec. III-C).

A. Internal Model Control

Internal Model Control (IMC) is a robust control framework that outperforms classical feedback control in handling model uncertainties and disturbances. Unlike feedback control, which corrects errors in real time, IMC uses a predictive model to anticipate and compensate for disturbances. Fig. 3 illustrates their structures, where P and $\bar{P}$ are the plant and internal models, Q is the IMC controller, and C is the classical feedback controller. The reference, disturbance, and output are denoted by r , d , and y , respectively. IMC computes the difference $\tilde{d} = \tilde{y} - y$ between predicted and actual responses, adjusting control inputs accordingly for enhanced robustness.

IMC is straightforward when the plant is easily modeled [3] but becomes challenging for nonlinear systems where approximations fail to capture true dynamics [30], [31]. This reliance on precise models limits its use in robotics, such as quadrotors with suspended payloads or legged robots on uneven terrain, where accurate system modeling is difficult. Our method integrates IMC with large-scale RL training to address these challenges. A simplified predictive model provides a rough yet stable estimate, while large-scale randomized simulations train a robust control policy that compensates for modeling inaccuracies. RL optimizes the policy with the predictive error feedback, revealing external disturbances and nonlinear dynamics, leading to more adaptive and effective control in real-world settings.

B. Design of a Simplified Predictive Model

The predictive model is key to our framework, enabling discrepancy detection and compensation. Its design faces two challenges: accurately estimating system response and closely aligning predictions with actual behavior. Precise estimation ensures the feedback loop reflects the true robot state, while alignment ensures reliability under varying conditions.

To streamline state estimation across diverse robots, we model only the rigid body state, excluding actuator- or limb-specific states. For robots equipped with monocular or binocular cameras and built-in IMUs, numerous visual(-inertial) odometry methods provide robust state estimations [32], [33], [34], [35], which can be used as the actual system response within the IMC structure. By treating the robot as a rigid body and using Euclidean transformation to estimate the next state, we significantly reduce the modeling complexity, making it more practical and efficient.

Here, we first present a general approach for formulating the predictive model and calculating the error, followed by specific adaptations for different robot morphologies in the next section. We assume that the model inputs include velocity or acceleration commands, and define the current body state as $\mathbf{x}_k = [\mathbf{p}_k, \mathbf{q}_k, \mathbf{v}_k, \boldsymbol{\omega}_k] \in \mathbb{R}^{13}$, where $\mathbf{p}_k$ is the position, $\mathbf{q}_k$ is the quaternion (orientation), $\mathbf{v}_k$ is the velocity, and $\boldsymbol{\omega}_k$ is the angular velocity. The time interval between two steps is denoted as Δt .

The next position and velocity of the body can be computed via Newton's law [36]:

$$\tilde{\mathbf{p}}_{k+1} = \mathbf{p}_k + \mathbf{v}_k \Delta t + \frac{1}{2} \mathbf{a}_k \Delta t^2, \quad (1)$$

$$\tilde{\mathbf{v}}_{k+1} = \mathbf{v}_k + \mathbf{a}_k \Delta t. \quad (2)$$

For rotation, we adopt a discrete update rule while handling singularities. First, we compute the change in angle $\Delta \theta$ with the angular velocity $\boldsymbol{\omega}_k$ and angular acceleration $\dot{\boldsymbol{\omega}}_k$:

$$\Delta \theta = \boldsymbol{\omega}_k \Delta t + \frac{1}{2} \dot{\boldsymbol{\omega}}_k \Delta t^2. \quad (3)$$

Next, we follow [37] to apply the exponential map to project the rotation from $\mathbb{R}^3$ to $SO(3)$, obtaining the corresponding quaternion $\mathbf{q}_{\Delta \theta}$:

$$\mathbf{q}_{\Delta \theta} = \begin{cases} \left[\cos\left(\frac{\|\Delta \theta\|}{2}\right), \frac{\sin\left(\frac{\|\Delta \theta\|}{2}\right)}{\|\Delta \theta\|} \cdot \Delta \theta \right], & \|\Delta \theta\| \leq \sqrt[4]{\epsilon_{\text{machine}}} \\ \left[\cos\left(\frac{\|\Delta \theta\|}{2}\right), \left(\frac{1}{2} + \frac{\|\Delta \theta\|^2}{48}\right) \cdot \Delta \theta \right], & \text{otherwise} \end{cases} \quad (4)$$

Finally, we can predict the next quaternion and angular velocity as follows, where $\otimes$ denotes quaternion multiplication:

$$\tilde{\mathbf{q}}_{k+1} = \mathbf{q}_k \otimes \mathbf{q}_{\Delta \theta} \quad (5)$$

$$\tilde{\boldsymbol{\omega}}_{k+1} = \boldsymbol{\omega}_k + \dot{\boldsymbol{\omega}}_k \Delta t \quad (6)$$

We have established an explicit method to predict the next body state, assuming the required motion can be represented by commands at the acceleration or velocity level. Then, given the predicted body state $\tilde{\mathbf{x}}_{k+1} = [\tilde{\mathbf{p}}_{k+1}, \tilde{\mathbf{q}}_{k+1}, \tilde{\mathbf{v}}_{k+1}, \tilde{\boldsymbol{\omega}}_{k+1}]$ and the actual body state $\mathbf{x}_{k+1}$, we compute the predictive error $\tilde{\mathbf{d}}_{k+1}$, which is then fed into the policy to provide effective feedback signals.

$$\tilde{\mathbf{d}}_{k+1} = [\Delta \tilde{\mathbf{p}}_{k+1}, 1 - \cos \tilde{\Theta}, \Delta \tilde{\mathbf{v}}_{k+1}, \Delta \tilde{\boldsymbol{\omega}}_{k+1}] \quad (7)$$

Here, $\Delta \tilde{\mathbf{p}}_{k+1}$, $\Delta \tilde{\mathbf{v}}_{k+1}$, and $\Delta \tilde{\boldsymbol{\omega}}_{k+1}$ represent the differences in position, velocity, and angular velocity, respectively. We use the cosine distance to quantify the difference between orientations, where $\tilde{\Theta}$ is the angle difference between the two quaternions.

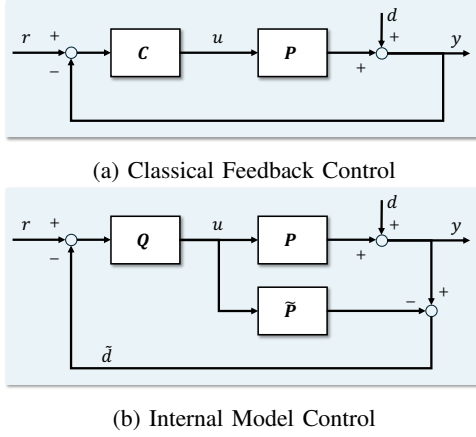


图 3: 经典反馈控制和内部模型控制 (IMC) 的反馈结构。

两种特定的机器人形态: 四旋翼和四足机器人 (第III-C节)。

A. Internal Model Control

内部模型控制 (IMC) 是一种稳健的控制框架, 在处理模型不确定性和扰动方面优于传统的反馈控制。与实时纠正误差的反馈控制不同, IMC 使用预测模型来预测并补偿扰动。图 3 说明了它们的结构, 其中 P 和 $\tilde{P}$ 分别是被控对象和内部模型, Q 是 IMC 控制器, C 是传统反馈控制器。参考信号、扰动和输出分别用 r 、 d 和 y 表示。IMC 计算预测响应与实际响应之间的差值 $\tilde{d} = \tilde{y} - y$, 并相应调整控制输入, 以增强稳健性。

当植物容易建模时, IMC 是直接的[3], 但对于近似无法捕捉真实动力学的非线性系统而言, 它变得具有挑战性[30], [31]。对精确模型的依赖限制了其在机器人领域的应用, 例如带悬挂载荷的四旋翼或在不平坦地形上行走的腿型机器人, 因为这些系统难以精确建模。我们的方法将 IMC 与大规模强化学习训练相结合, 以应对这些挑战。简化的预测模型提供粗略但稳定的估计, 而大规模随机模拟训练出能够弥补建模不准确性的稳健控制策略。强化学习通过预测误差反馈优化策略, 从而揭示外部干扰和非线性动力学, 使在现实环境中实现更加自适应和有效的控制成为可能。

B. Design of a Simplified Predictive Model

预测模型是我们框架的关键, 使得差异检测和补偿成为可能。它的设计面临两个挑战: 准确估计系统响应和使预测与实际行为紧密一致。精确的估计确保反馈环能够反映机器人的真实状态, 而一致性则确保在各种条件下的可靠性。

为了简化不同机器人间的状态估计, 我们仅建模刚体状态, 不包括执行器或肢体的特定状态。对于配备了单目或双目相机以及内置惯性测量单元 (IMU) 的机器人, 许多视觉 (-惯性) 里程计方法能够提供稳健的状态估计 [32], [33], [34], [35], 这些可以作为 IMC 结构中的实际系统响应。通过将机器人视为刚体并使用欧几里得变换来估计下一状态, 我们大幅降低了建模复杂性, 使其更实用和高效。

在这里, 我们首先提出一种用于构建预测模型和计算误差的一般方法, 接着在下一节中针对不同的机器人形态进行具体调整。我们假设模型的输入包括速度或加速度指令, 并将当前的机体状态定义为 $\mathbf{x}_k = [\mathbf{p}_k, \mathbf{q}_k, \mathbf{v}_k, \boldsymbol{\omega}_k] \in \mathbb{R}^{13}$, 其中 $\mathbf{p}_k$ 为位置, $\mathbf{q}_k$ 为四元数 (姿态), $\mathbf{v}_k$ 为速度, $\boldsymbol{\omega}_k$ 为角速度。两个步骤之间的时间间隔记作 Δt 。

物体的下一个位置和速度可以通过牛顿定律 [36] 计算:

$$\tilde{\mathbf{p}}_{k+1} = \mathbf{p}_k + \mathbf{v}_k \Delta t + \frac{1}{2} \mathbf{a}_k \Delta t^2, \quad (1)$$

$$\tilde{\mathbf{v}}_{k+1} = \mathbf{v}_k + \mathbf{a}_k \Delta t. \quad (2)$$

对于旋转, 我们在处理奇异性时采用离散更新规则。首先, 我们用角速度 $\boldsymbol{\omega}_k$ 和角加速度 $\dot{\boldsymbol{\omega}}_k$ 计算角度变化 $\Delta \boldsymbol{\theta}$:

$$\Delta \boldsymbol{\theta} = \boldsymbol{\omega}_k \Delta t + \frac{1}{2} \dot{\boldsymbol{\omega}}_k \Delta t^2. \quad (3)$$

接下来, 我们参考 [37] 对指数映射进行应用, 将旋转从 $\mathbb{R}^3$ 投影到 $SO(3)$, 得到相应的四元数 $\mathbf{q}_{\Delta \boldsymbol{\theta}}$:

$$\mathbf{q}_{\Delta \boldsymbol{\theta}} = \begin{cases} \left[\cos\left(\frac{\|\Delta \boldsymbol{\theta}\|}{2}\right), \frac{\sin\left(\frac{\|\Delta \boldsymbol{\theta}\|}{2}\right)}{\|\Delta \boldsymbol{\theta}\|} \cdot \Delta \boldsymbol{\theta} \right], & \|\Delta \boldsymbol{\theta}\| \leq \sqrt[4]{\epsilon_{\text{machine}}} \\ \left[\cos\left(\frac{\|\Delta \boldsymbol{\theta}\|}{2}\right), \left(\frac{1}{2} + \frac{\|\Delta \boldsymbol{\theta}\|^2}{48}\right) \cdot \Delta \boldsymbol{\theta} \right], & \text{otherwise} \end{cases} \quad (4)$$

最后, 我们可以如下预测下一个四元数和角速度, 其中 $\otimes$ 表示四元数乘法:

$$\tilde{\mathbf{q}}_{k+1} = \mathbf{q}_k \otimes \mathbf{q}_{\Delta \boldsymbol{\theta}} \quad (5)$$

$$\tilde{\boldsymbol{\omega}}_{k+1} = \boldsymbol{\omega}_k + \dot{\boldsymbol{\omega}}_k \Delta t \quad (6)$$

我们已经建立了一种明确的方法来预测下一个身体状态, 假设所需的运动可以通过加速度或速度级别的指令来表示。然后, 给定预测的身体状态 $\tilde{\mathbf{x}}_{k+1} = [\tilde{\mathbf{p}}_{k+1}, \tilde{\mathbf{q}}_{k+1}, \tilde{\mathbf{v}}_{k+1}, \tilde{\boldsymbol{\omega}}_{k+1}]$ 和实际的身体状态 $\mathbf{x}_{k+1}$, 我们计算预测误差 $\tilde{\mathbf{d}}_{k+1}$, 然后将其输入策略中以提供有效的反馈信号。

$$\tilde{\mathbf{d}}_{k+1} = [\Delta \tilde{\mathbf{p}}_{k+1}, 1 - \cos \tilde{\Theta}, \Delta \tilde{\mathbf{v}}_{k+1}, \Delta \tilde{\boldsymbol{\omega}}_{k+1}] \quad (7)$$

这里, $\Delta \tilde{\mathbf{p}}_{k+1}$ 、 $\Delta \tilde{\mathbf{v}}_{k+1}$ 和 $\Delta \tilde{\boldsymbol{\omega}}_{k+1}$ 分别表示位置、速度和角速度的差异。我们使用余弦距离来量化方向的差异, 其中 $\tilde{\Theta}$ 是两个四元数之间的角度差异。

TABLE I: Parameter ranges for dynamics randomization on quadrotor. $K_p^0 = [37, 37, 11]$

Parameter Name	Train Range	Eval Range
Mass (M)	[0.142, 0.950]	[0.114, 1.140]
Arm Length (L)	[0.046, 0.200]	[0.037, 0.240]
Width-to- L Ratio (α_w)	[1.0, 1.414]	[0.852, 1.732]
Height-to- L Ratio (α_h)	[0.577, 1.0]	[0.268, 1.414]
Motor Force Constant (K_f)	[1.15e-7, 7.64e-6]	[9.16e-8, 9.17e-6]
Max Thrust-to-Weight Ratio	[2.0, 3.5]	[1.6, 4.2]
Motor Drag Constant (κ)	[0.0041, 0.0168]	[0.0033, 0.0201]
Motor Time Constant (τ_c)	[0.3, 0.5]	[0.24, 0.6]
P Gains (K_p)	$[-0.3, 0.3] \times K_p^0$	$[-0.5, 0.5] \times K_p^0$

C. Specifications for Common Robots

In this section, we further explain how to derive velocity or acceleration commands for various robots, including quadrotors and quadrupedal robots. We demonstrate how these commands align with existing control methods, ensuring seamless integration into the policy learning process.

1) *Quadrotors*: Traditional quadrotor controllers use cascaded structures [38], allowing flexible integration of neural controllers. Recent studies show that using collective thrust and bodyrate (CTBR) as the action space enhances resilience to dynamics mismatches, improves domain transfer, and boosts agility [39], [20], [9]. This design aligns with our framework: the policy’s bodyrate command updates rotation, while collective thrust determines linear acceleration. Consequently, predictive errors, computed from prior policy output and successive body states, integrate directly into the current policy input without modifying the learning algorithm.

2) *Quadrupedal robots*: Unlike quadrotors, legged robots, particularly quadrupeds and bipeds, are well-suited for RL-based control due to their high-dimensional actuation and terrain interactions [8], [11], [12], [14], [19]. Neural controllers typically generate actuator targets to achieve desired body velocities (v_x, v_y, ω_z). To integrate our predictive model into the control loop, we use reference velocity commands as model inputs, deriving linear and angular accelerations by differentiating consecutive commands. Setting other accelerations to zero ensures seamless integration with existing policies. This method is compatible with other algorithms via a plug-and-play feedback loop augmenting policy inputs.

IV. EXPERIMENTAL RESULTS

To evaluate our framework’s performance and versatility, we conduct experiments on quadrotors and quadrupedal robots, comparing against state-of-the-art methods in each domain. Then, we perform ablation studies on the feedback structure and noise tolerance of the simplified predictive model. Finally, we present sim-to-real deployment results on a quadrotor with two rope-suspended payloads, showcasing its effectiveness in enhancing system robustness.

A. Experiment Settings

1) *Quadrotors*: For quadrotors, controllers are trained using a customized simulator based on the dynamics shown in [40], where the collective thrust and bodyrate commands are

followed by a PD controller. The fourth-order Runge-Kutta method [41] updates quadrotor states at each simulation step with a update frequency at 200 Hz, and the controller’s execution frequency is 50 Hz. Following Huang et al. [20], we train controllers on two types of reference trajectories, including smooth chained polynomial trajectories and infeasible zigzag trajectories. External disturbances are applied as randomized force perturbations within $[-3.3 \text{ m/s}^2, 3.3 \text{ m/s}^2]$, representing Brownian motion variations as in [20]. Dynamical parameters are randomized at the start of each episode for training, and unseen parameters are used for evaluation to test robustness. The detailed randomization range is shown in Tab. I. We train and evaluate each method using five random seeds, reporting the average performance across 1,024 parallel trials. For evaluation, we test controllers trained with smooth trajectories on two types of randomly generated smooth paths: circle and chained polynomial, while controllers trained on infeasible paths are tested on 5-point star and zigzag trajectories.

2) *Quadrupedal robots*: Following Long et al. [13], we train controllers for the Unitree Aliengo using Isaac Gym [17] with 4096 parallel environments and a 100-step rollout. The training curriculum includes progressively challenging terrains and reference commands. Controllers are trained on terrains such as pyramid slopes, rough pyramid slopes, stairs, and flats with discrete obstacles, with distributions of [0.1, 0.2, 0.6, 0.1]. We evaluate methods by tracking x-y linear velocity or yaw rate commands under two conditions: no disturbances and random force pushes. The x-y linear velocity command is sampled from $[-1.0, 1.0] \text{ m/s}$, and the yaw rate command from $[-1.0, 1.0] \text{ rad/s}$. During linear velocity tracking evaluations, the angular velocity target is set to zero, and vice versa for angular velocity tracking. For linear velocity tracking, the forward and lateral velocities are uniformly sampled from $[-1 \text{ m/s}, 1 \text{ m/s}]$. For angular velocity tracking tasks, the yaw rate is uniformly sampled from $[-1 \text{ rad/s}, 1 \text{ rad/s}]$. The tracking error is averaged over five random seeds, each with 4096 parallel environments. To validate robustness, we emulate an impulse push by setting a randomized base velocity to the robot every 16 steps, uniformly sampled within the maximum range. The maximum impulse velocity during training is 1 m/s, increased to 1.5 m/s for evaluation.

B. Experiments on Quadrotors

We compare our method for quadrotors against two classical control methods ($\mathcal{L}_1$ -PID and $\mathcal{L}_1$ -MPPI) and four learning-based baselines (CAL [28], PPO [27], RMA [12], DATT [20]), using two variants each for RMA and DATT to enable deeper analyses. All methods use the same PPO training parameters. For CAL [28], a sum combination of MPPI and RL, MPPI serves as a sampling-based non-linear MPC method, with the RL policy compensating the MPPI output. We set the sampling number to 4096 and the rollout horizon to 20 in MPPI’s sampling procedure. PPO employs a three-layer MLP policy and a temporal convolutional network (TCN) to process future waypoints in the reference trajectory, which is inherited by all other methods. In addition, RMA adopts a teacher-student training framework, with RMA(T) as the teacher model and

表 I: 四旋翼动力学随机化的参数范围。 $K_p^0 = [37, 37, 11]$

Parameter Name	Train Range	Eval Range
Mass (M)	[0.142, 0.950]	[0.114, 1.140]
Arm Length (L)	[0.046, 0.200]	[0.037, 0.240]
Width-to- L Ratio (α_w)	[1.0, 1.414]	[0.852, 1.732]
Height-to- L Ratio (α_h)	[0.577, 1.0]	[0.268, 1.414]
Motor Force Constant (K_f)	[1.15e-7, 7.64e-6]	[9.16e-8, 9.17e-6]
Max Thrust-to-Weight Ratio	[2.0, 3.5]	[1.6, 4.2]
Motor Drag Constant (κ)	[0.0041, 0.0168]	[0.0033, 0.0201]
Motor Time Constant (τ_c)	[0.3, 0.5]	[0.24, 0.6]
P Gains (K_p)	$[-0.3, 0.3] \times K_p^0$	$[-0.5, 0.5] \times K_p^0$

C. Specifications for Common Robots

在本节中，我们将进一步解释如何为各种机器人推导速度或加速度指令，包括四旋翼和四足机器人。我们演示了这些指令如何与现有控制方法对齐，确保能够无缝地整合到策略学习过程中。

1) *Quadrotors*: 传统的四旋翼控制器使用级联结构 [38]，允许神经控制器的灵活整合。最近的研究表明，使用总推力和机体角速率 (CTBR) 作为动作空间可以增强对动态不匹配的鲁棒性，提高领域迁移能力，并提升机动性 [39], [20], [9]。这一设计与我们的框架保持一致：策略的机体角速率指令更新旋转，而总推力决定线性加速度。因此，由之前策略输出和连续机体状态计算的预测误差可以直接整合到当前策略输入中，而无需修改学习算法。

2) *Quadrupedal robots*: 与四旋翼不同，有腿机器人，尤其是四足和两足机器人，由于其高维的驱动能力和地形交互，非常适合基于强化学习的控制 [8], [11], [12], [14], [19]。神经控制器通常生成执行器目标以实现期望的身体速度 (v_x, v_y, ω_z)。为了将我们的预测模型集成到控制回路中，我们使用参考速度指令作为模型输入，通过对连续指令进行微分来得出线性和角加速度。将其他加速度设为零可以确保与现有策略的无缝集成。这种方法通过即插即用的反馈回路增强策略输入，与其他算法兼容。

IV. 实验结果

为了评估我们框架的性能和多样性，我们在四旋翼和四足机器人上进行了实验，并与各领域的最先进方法进行了比较。随后，我们对简化预测模型的反馈结构和抗噪能力进行了消融研究。最后，我们展示了在搭载两个绳索悬挂负载的四旋翼上的模拟到真实部署结果，展示了其在提升系统鲁棒性方面的有效性。

A. Experiment Settings

1) *Quadrotors*: 对于四旋翼，控制器是使用基于[40]中所示动力学的定制模拟器进行训练的，其中包括总推力和机体角速度指令

随后是一个 PD 控制器。四阶 Runge-Kutta 方法 [41] 在每个仿真步骤更新四旋翼状态，更新频率为 200 Hz，控制器的执行频率为 50 Hz。按照 Huang 等人 [20] 的方法，我们在两种类型的参考轨迹上训练控制器，包括平滑链式多项式轨迹和不可实现的之字形轨迹。外部扰动以随机力扰动的形式施加，范围在 $[-3.3 \text{ m/s}^2, 3.3 \text{ m/s}^2]$ 内，表示如 [20] 中的布朗运动变化。训练时，每个回合开始时随机化动力学参数，评估时使用未见过的参数以测试鲁棒性。详细的随机化范围如表 I 所示。我们使用五个随机种子训练和评估每种方法，并报告 1,024 次并行试验的平均性能。评估时，我们测试在平滑轨迹上训练的控制器在两种随机生成的平滑路径上：圆形和

2) *Quadrupedal robots*: 遵循 Long 等人 [13] 的方法，我们使用 Isaac Gym [17] 对 Unitree Aliengo 进行控制器训练，采用 4096 个并行环境和 100 步 rollout。训练课程包括逐步增加难度的地形和参考指令。控制器在金字塔斜坡、粗糙金字塔斜坡、楼梯和平坦带离散障碍物的地形上进行训练，分布分别为 [0.1, 0.2, 0.6, 0.1]。我们通过跟踪 x-y 线速度或偏航速率指令，在两种条件下评估方法：无干扰和随机力推。x-y 线速度指令取自 $[-1.0, 1.0] \text{ m/s}$ ，而偏航速率指令取自 $[-1.0, 1.0] \text{ rad/s}$ 。在线速度跟踪评估中，角速度目标设置为零；而在角速度跟踪评估中则相反。在线速度跟踪任务中，前向和横向速度均从 $[-1 \text{ m/s}, 1 \text{ m/s}]$ 均匀取样。对于角速度跟踪任务，偏航速率为未

B. Experiments on Quadrotors

我们将我们的方法在四旋翼上的表现与两种经典控制方法 ($\mathcal{L}_1$ -PID 和 $\mathcal{L}_1$ -MPPI) 以及四种基于学习的基线方法 (CAL [28]、PPO [27]、RMA [12]、DATT [20]) 进行了比较，并对 RMA 和 DATT 各使用两种变体以进行更深入的分析。所有方法使用相同的 PPO 训练参数。对于 CAL [28] (MPPI 与 RL 的总和组合)，MPPI 作为一种基于采样的非线性 MPC 方法，RL 策略用来补偿 MPPI 的输出。我们将 MPPI 采样过程中的采样数量设置为 4096，展开时域设置为 20。PPO 使用一个三层 MLP 策略和一个时间卷积网络 (TCN) 来处理参考轨迹中的未来航点，该结构被所有其他方法继承。此外，RMA 采用教师-学生训练框架，其中 RMA(T) 作为教师模型，而

TABLE II: Simulation results for tracking errors (unit: m) with quadrotors, averaged over five random seeds. **Bold values** denote the best results. **Blue cells** indicate performance under random external force disturbances, with percentages in brackets representing the relative change in error compared to static conditions. T and S refer to the teacher and student models for RMA, respectively, while DATT with GT indicates the use of ground-truth disturbance inputs instead of $\mathcal{L}_1$ estimates.

Config	Method	Smooth Trajectory				Infeasible Trajectory			
		Circle		Chained Poly		5-Point Star		Zigzag	
Train	$\mathcal{L}_1$ -PID	0.214	0.237 (+10.74%)	0.043	0.054 (+25.58%)	<i>crashed</i>		<i>crashed</i>	
	$\mathcal{L}_1$ -MPPI	0.202	0.215 (+6.44%)	0.092	0.117 (+27.17%)	0.193	0.216 (+11.91%)	0.169	0.186 (+10.06%)
	CAL	0.087	0.157 (+80.46%)	0.056	0.118 (+110.71%)	0.103	0.175 (+69.90%)	0.105	0.164 (+56.19%)
	PPO	0.102	0.169 (+65.69%)	0.065	0.117 (+80.00%)	0.097	0.172 (+77.32%)	0.095	0.154 (+62.11%)
	RMA (T)	0.071	0.080 (+12.68%)	0.043	0.047 (+9.30%)	0.087	0.094 (+8.05%)	0.074	0.082 (+10.81%)
	RMA (S)	0.084	0.110 (+30.95%)	0.048	0.064 (+33.33%)	0.116	0.157 (+35.34%)	0.112	0.142 (+26.79%)
	DATT ($\mathcal{L}_1$)	0.084	0.128 (+52.38%)	0.051	0.076 (+49.02%)	0.089	0.163 (+83.15%)	0.077	0.138 (+79.22%)
	DATT (GT)	0.084	0.093 (+10.71%)	0.052	0.055 (+5.77%)	0.089	0.098 (+10.11%)	0.077	0.084 (+9.09%)
	Ours	0.072	0.083 (+15.28%)	0.032	0.038 (+18.75%)	0.065	0.073 (+12.31%)	0.057	0.064 (+12.28%)
Eval	$\mathcal{L}_1$ -PID	0.372	0.416 (+11.82%)	0.061	0.107 (+75.41%)	<i>crashed</i>		<i>crashed</i>	
	$\mathcal{L}_1$ -MPPI	0.212	0.234 (+10.38%)	0.097	0.132 (+36.08%)	0.207	0.253 (+22.22%)	0.176	0.218 (+23.86%)
	CAL	0.224	0.321 (+43.30%)	0.171	0.251 (+46.78%)	0.262	0.364 (+38.93%)	0.265	0.358 (+35.09%)
	PPO	0.265	0.351 (+32.45%)	0.183	0.273 (+49.18%)	0.257	0.374 (+45.53%)	0.265	0.367 (+38.49%)
	RMA (T)	0.117	0.147 (+25.64%)	0.072	0.097 (+34.72%)	0.151	0.192 (+27.15%)	0.145	0.187 (+28.97%)
	RMA (S)	0.280	0.360 (+28.57%)	0.174	0.243 (+39.66%)	0.358	0.424 (+18.44%)	0.349	0.417 (+19.48%)
	DATT ($\mathcal{L}_1$)	0.137	0.523 (+281.75%)	0.094	0.354 (+276.60%)	0.169	0.585 (+246.15%)	0.160	0.559 (+249.38%)
	DATT (GT)	0.139	0.167 (+20.14%)	0.094	0.119 (+26.60%)	0.178	0.197 (+10.67%)	0.156	0.189 (+21.15%)
	Ours	0.113	0.135 (+19.47%)	0.054	0.073 (+35.19%)	0.103	0.134 (+30.10%)	0.101	0.132 (+30.69%)

RMA(S) as the student. RMA(T) integrates dynamic parameters and ground-truth force disturbances, while RMA(S) uses a TCN to approximate the teacher’s latent state from state-action histories. In contrast, DATT utilizes an $\mathcal{L}_1$ disturbance estimator and tracking error inputs to improve trajectory-tracking robustness. To separate the $\mathcal{L}_1$ estimator’s effect, we also include a privileged variant, DATT(GT), which directly uses ground-truth force disturbances in place of the estimator.

As shown in Tab. II, our method consistently outperforms others across nearly all tasks, particularly under disturbances and out-of-distribution (OOD) system dynamics.

Among estimator-based baselines—both explicit (e.g., $\mathcal{L}_1$ estimator in $\mathcal{L}_1$ -PID, $\mathcal{L}_1$ -MPPI, DATT) and implicit (e.g., RMA’s neural encoder)—DATT is the most resilient, benefiting from its explicit $\mathcal{L}_1$ estimator coupled with an RL-based controller. Nevertheless, all estimator-based methods degrade sharply under our extreme out-of-distribution scenarios—where internal dynamics shift or sudden external forces occur—because their estimators become unreliable and control robustness is compromised. Estimator-free approaches such as CAL and PPO also fail to manage these varied disturbances. In contrast, our method computes the predictive error between expected and observed body transformations, directly signaling state drift and disturbance magnitude. A policy trained on this error maintains strong performance across both nominal and unexpected conditions.

We also acknowledge that limited hyperparameter sweeps in our experiments may contribute to the baselines’ degraded performance. To ensure a fair comparison, we include privileged variants (DATT (GT) and RMA (T)) that receive ground-truth states rather than estimates, representing an upper tuning bound. Even against these privileged methods, our predictive error-driven controller achieves superior and more stable control, supporting our hypothesis that explicit predictive error feedback offers a simple, generalizable path to robust

performance.

C. Experiments on Quadrupedal Robots

For quadrupedal robots, we compare our methods with PPO [27], RMA [12] and HIM [13]. PPO and RMA serve as standard and common baselines in continuous control for quadrupeds, while HIM leverages unsupervised contrastive learning to train a predictive encoder for the next latent state and body velocity. Here, we directly use the teacher-policy of RMA to set a strong baseline. Our method enhances the PPO policy by integrating predictive error feedback. Additionally, we evaluate a variant of our method, adding predictive error feedback to both the inputs of the HIM policy and its predictive encoder during training and evaluation.

As shown in Tab. III, our method outperforms HIM by 26–44% across various test conditions and matches or surpasses the privileged RMA. The combination of our framework with HIM also shows improvements over the original HIM. Interestingly, the predictive encoder in HIM shows a slight degradation compared to standard PPO when combined with our NeuralIMC structure. We hypothesize that this is because the predictive error provides sufficient signal for robust control, rendering the additional impact of predictive encoding negligible. This suggests that predictive error feedback alone is sufficiently informative for enhancing control robustness, simplifying the neural control pipeline, and facilitating integration with other learning algorithms.

D. Ablation Studies

In this section, we conduct three ablation studies to analyze our framework. Inspired by [14], we first examine whether incorporating short- and long-horizon input/output histories improves performance. Next, we introduce noise into the predictive model’s input and output to assess its robustness and the impact of prediction accuracy on control. Finally, we

表 II: 四旋翼跟踪误差的仿真结果 (单位: 米), 取五个随机种子的平均值。加粗数值表示最佳结果。蓝色单元格表示在随机外力扰动下的性能, 括号中的百分比表示相对于静态条件的误差相对变化。T 和 S 分别指 RMA 的教师模型和学生模型, 而带 GT 的 DATT 表示使用真实扰动输入而非 $\mathcal{L}_1$ 估计值。

Config	Method	Smooth Trajectory				Infeasible Trajectory			
		Circle		Chained Poly		5-Point Star		Zigzag	
Train	$\mathcal{L}_1$ -PID	0.214	0.237 (+10.74%)	0.043	0.054 (+25.58%)	<i>crashed</i>		<i>crashed</i>	
	$\mathcal{L}_1$ -MPPI	0.202	0.215 (+6.44%)	0.092	0.117 (+27.17%)	0.193	0.216 (+11.91%)	0.169	0.186 (+10.06%)
	CAL	0.087	0.157 (+80.46%)	0.056	0.118 (+110.71%)	0.103	0.175 (+69.90%)	0.105	0.164 (+56.19%)
	PPO	0.102	0.169 (+65.69%)	0.065	0.117 (+80.00%)	0.097	0.172 (+77.32%)	0.095	0.154 (+62.11%)
	RMA (T)	0.071	0.080 (+12.68%)	0.043	0.047 (+9.30%)	0.087	0.094 (+8.05%)	0.074	0.082 (+10.81%)
	RMA (S)	0.084	0.110 (+30.95%)	0.048	0.064 (+33.33%)	0.116	0.157 (+35.34%)	0.112	0.142 (+26.79%)
	DATT ($\mathcal{L}_1$)	0.084	0.128 (+52.38%)	0.051	0.076 (+49.02%)	0.089	0.163 (+83.15%)	0.077	0.138 (+79.22%)
	DATT (GT)	0.084	0.093 (+10.71%)	0.052	0.055 (+5.77%)	0.089	0.098 (+10.11%)	0.077	0.084 (+9.09%)
	Ours	0.072	0.083 (+15.28%)	0.032	0.038 (+18.75%)	0.065	0.073 (+12.31%)	0.057	0.064 (+12.28%)
Eval	$\mathcal{L}_1$ -PID	0.372	0.416 (+11.82%)	0.061	0.107 (+75.41%)	<i>crashed</i>		<i>crashed</i>	
	$\mathcal{L}_1$ -MPPI	0.212	0.234 (+10.38%)	0.097	0.132 (+36.08%)	0.207	0.253 (+22.22%)	0.176	0.218 (+23.86%)
	CAL	0.224	0.321 (+43.30%)	0.171	0.251 (+46.78%)	0.262	0.364 (+38.93%)	0.265	0.358 (+35.09%)
	PPO	0.265	0.351 (+32.45%)	0.183	0.273 (+49.18%)	0.257	0.374 (+45.53%)	0.265	0.367 (+38.49%)
	RMA (T)	0.117	0.147 (+25.64%)	0.072	0.097 (+34.72%)	0.151	0.192 (+27.15%)	0.145	0.187 (+28.97%)
	RMA (S)	0.280	0.360 (+28.57%)	0.174	0.243 (+39.66%)	0.358	0.424 (+18.44%)	0.349	0.417 (+19.48%)
	DATT ($\mathcal{L}_1$)	0.137	0.523 (+281.75%)	0.094	0.354 (+276.60%)	0.169	0.585 (+246.15%)	0.160	0.559 (+249.38%)
	DATT (GT)	0.139	0.167 (+20.14%)	0.094	0.119 (+26.60%)	0.178	0.197 (+10.67%)	0.156	0.189 (+21.15%)
	Ours	0.113	0.135 (+19.47%)	0.054	0.073 (+35.19%)	0.103	0.134 (+30.10%)	0.101	0.132 (+30.69%)

RMA(S) 作为学生。RMA(T) 集成了动态参数和真实力扰动, 而 RMA(S) 使用 TCN 从状态-动作历史中近似教师的潜在状态。相比之下, DATT 利用 $\mathcal{L}_1$ 扰动估计器和跟踪误差输入来提高轨迹跟踪的鲁棒性。为了分离 $\mathcal{L}_1$ 估计器的影响, 我们还包括一个特权变体 DATT(GT), 该变体直接使用真实力扰动替代估计器。

如表 II 所示, 我们的方法在几乎所有任务中始终优于其他方法, 尤其是在干扰和分布外 (OOD) 系统动态下。

在基于估计器的基线方法中——包括显式方法 (例如 $\mathcal{L}_1$ -PID 中的 $\mathcal{L}_1$ 估计器、 $\mathcal{L}_1$ -MPPI、DATT) 和隐式方法 (例如 RMA 的神经编码器) ——DATT 最具弹性, 得益于其显式的 $\mathcal{L}_1$ 估计器结合基于强化学习的控制器。然而, 在我们的极端分布外场景下——内部动力学发生变化或出现突发外力——所有基于估计器的方法都会急剧下降, 因为它们的估计器变得不可靠, 控制的鲁棒性受到影响。无估计器的方法, 如 CAL 和 PPO, 也无法应对这些多样的干扰。相比之下, 我们的方法计算期望与观察到的身体变换之间的预测误差, 直接反映状态漂移和干扰强度。基于该误差训练的策略在标称和意外条件下均能保持强劲的性能。

我们也承认, 在我们的实验中有限的超参数搜索可能导致基线方法性能下降。为确保公平比较, 我们包括了特权变体 (DATT (GT) 和 RMA (T)), 它们使用真实状态而非估计值, 代表了一个调优的上限。即使面对这些特权方法, 我们的预测误差驱动控制器仍实现了更优且更稳定的控制, 这支持了我们的假设: 明确的预测误差反馈提供了一条简单且可推广的方式来实现稳健性。

性能。

C. Experiments on Quadrupedal Robots

对于四足机器人, 我们将我们的方法与 PPO [27]、RMA [12] 和 HIM [13] 进行比较。PPO 和 RMA 作为四足连续控制中的标准且常用的基线方法, 而 HIM 利用无监督对比学习训练预测编码器, 以预测下一个潜在状态和身体速度。在这里, 我们直接使用 RMA 的教师策略来设定一个强基线。我们的方法通过整合预测误差反馈来增强 PPO 策略。此外, 我们还评估了我们方法的一个变体, 在训练和评估期间, 将预测误差反馈添加到 HIM 策略的输入和其预测编码器中。

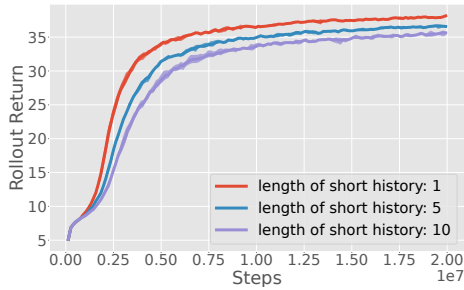
如表 III 所示, 我们的方法在各种测试条件下比 HIM 高出 26–44%, 并且与特权 RMA 相媲美或超越。将我们的框架与 HIM 结合也显示出比原始 HIM 更好的改进。有趣的是, 当 HIM 中的预测编码器与我们的 NeuralIMC 结构结合时, 相比标准 PPO 表现出轻微的退化。我们推测, 这是因为预测误差提供了足够的信号以实现稳健控制, 从而使预测编码的额外影响可以忽略不计。这表明, 仅预测误差反馈就足够提供信息以增强控制稳健性, 简化神经控制流程, 并促进与其他学习算法的集成。

D. Ablation Studies

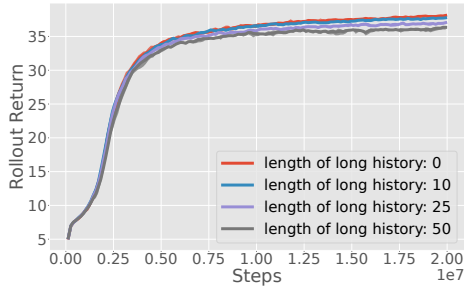
在本节中, 我们进行了三个消融实验来分析我们的框架。受到 [14] 的启发, 我们首先检查是否结合短期和长期的输入/输出历史能够提高性能。接下来, 我们在预测模型的输入和输出中引入噪声, 以评估其鲁棒性以及预测精度对控制的影响。最后, 我们

TABLE III: Simulation results on quadrupedal robots, averaged over five random seeds. The tracking error for linear velocity and angular velocity are computed as $\|v_{x,y} - v_{x,y}^{\text{target}}\|_2$ (unit: m/s) and $\|\omega_{\text{yaw}} - \omega_{\text{yaw}}^{\text{target}}\|_2$ (unit: rad/s), respectively. The best results for each case are highlighted in **bold**.

Disturbance	Method	Slopes		Rough Slopes		Stairs		Discrete Obstacles	
		Linear	Angular	Linear	Angular	Linear	Angular	Linear	Angular
None	PPO	0.132	0.085	0.143	0.087	0.200	0.085	0.163	0.085
	RMA	0.084	0.050	0.086	0.051	0.127	0.050	0.110	0.050
	HIM	0.100	0.067	0.103	0.068	0.181	0.067	0.147	0.067
	Ours (w/ HIM)	0.077	0.052	0.079	0.053	0.126	0.053	0.105	0.052
	Ours	0.071	0.048	0.073	0.049	0.101	0.048	0.085	0.048
Impulse	PPO	0.151	0.101	0.162	0.103	0.216	0.101	0.178	0.101
	RMA	0.096	0.059	0.098	0.060	0.137	0.060	0.120	0.059
	HIM	0.117	0.080	0.120	0.081	0.186	0.080	0.157	0.080
	Ours (w/ HIM)	0.091	0.061	0.093	0.061	0.139	0.061	0.117	0.061
	Ours	0.086	0.058	0.087	0.059	0.115	0.059	0.099	0.058



(a) Varying short-history length.



(b) Varying long-history length.

Fig. 4: Ablation study on the feedback structure with different history lengths.

compare our method using the predictive model versus a full-dynamics model to evaluate the effect of model simplification. All experiments involve quadrotors tracking zigzag trajectories across three random seeds.

1) *History structure*: We evaluate feedback structures incorporating historical records. In short-history experiments, we concatenate recent input/output pairs, including predictive errors, into a single input vector. For long-history experiments, we retain the latest state-action pair while encoding additional history using a TCN model [12], [14], concatenating its latent output with the policy input.

As shown in Fig. 4, using longer history—either short- or long-term—does not improve performance over immediate feedback (i.e., the latest one-step history). While differing from [14], this aligns with $\mathcal{L}_1$ adaptive control [21] and robust control methods [2], [3], where immediate error feedback suf-

TABLE IV: Evaluation of the internal predictive model’s noise tolerance, tested on quadrotors tracking zigzag trajectories.

	Noise σ_n	Train		Eval	
		static	disturbed	static	disturbed
	0.	0.057	0.064 (+12.28%)	0.101	0.119 (+17.82%)
input noise	0.02	0.063	0.069 (+9.52%)	0.092	0.117 (+27.17%)
	0.04	0.070	0.094 (+34.28%)	0.100	0.154 (+54.00%)
	0.08	0.071	0.121 (+70.42%)	0.134	0.202 (+50.74%)
	0.16	0.082	0.135 (+64.63%)	0.152	0.231 (+51.97%)
output noise	0.02	0.063	0.071 (+12.69%)	0.076	0.132 (+73.68%)
	0.04	0.073	0.097 (+32.87%)	0.102	0.161 (+57.84%)
	0.08	0.074	0.121 (+63.51%)	0.118	0.179 (+51.69%)
	0.16	0.076	0.130 (+71.05%)	0.138	0.222 (+60.86%)

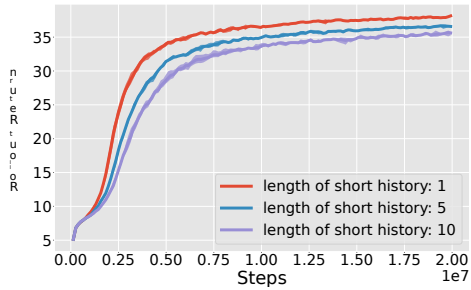
fices for accuracy and robustness. This simpler structure also reduces implementation complexity and mitigates cumulative errors in real robots.

2) *Noise tolerance*: To assess the impact of the internal predictive model’s accuracy on control performance, we introduce varying levels of noise into the model’s state input or predictions. Specifically, we introduce zero-mean Gaussian noise with varying variances to the state vector, including position, velocity, quaternion, and angular velocity. For the quaternion, we treat the sampled Gaussian noise as a delta Euler angle, convert it to a delta quaternion, and apply it to rotate the original orientation. We train and evaluate multiple policies using a range of noisy predictive models for quadrotor tracking on zigzag trajectories, with the results presented in Tab. IV.

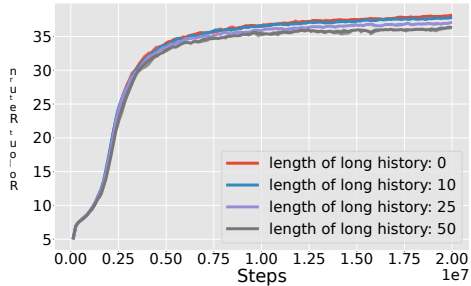
The results show that our method maintains competitive performance under mild noise, whether applied to model inputs or responses. However, as noise levels increase, the accuracy of the predictive model becomes increasingly crucial, particularly in scenarios with external disturbances or out-of-distribution system dynamics. Across different training and evaluation ranges, performance declines as noise intensifies, with a similar trend observed under external force disturbances. This observation highlights an advantage of our explicit predictive model over other learning-based approaches: it provides a stable baseline estimation, albeit approximate. Although modeling noise can affect feedback reliability, the rigid-body transformation adheres to fundamental motion principles and generalizes across robotic systems. While this model simplifies

表 III: 四足机器人仿真结果, 取五个随机种子的平均值。线速度和角速度的跟踪误差分别计算为 $\|v_{x,y} - v_{x,y}^{\text{target}}\|_2$ (单位: 米/秒) 和 $\|\omega_{\text{yaw}} - \omega_{\text{yaw}}^{\text{target}}\|_2$ (单位: 弧度/秒)。每种情况下的最佳结果以加粗显示。

Disturbance	Method	Slopes		Rough Slopes		Stairs		Discrete Obstacles	
		Linear	Angular	Linear	Angular	Linear	Angular	Linear	Angular
None	PPO	0.132	0.085	0.143	0.087	0.200	0.085	0.163	0.085
	RMA	0.084	0.050	0.086	0.051	0.127	0.050	0.110	0.050
	HIM	0.100	0.067	0.103	0.068	0.181	0.067	0.147	0.067
	Ours (w/ HIM)	0.077	0.052	0.079	0.053	0.126	0.053	0.105	0.052
	Ours	0.071	0.048	0.073	0.049	0.101	0.048	0.085	0.048
Impulse	PPO	0.151	0.101	0.162	0.103	0.216	0.101	0.178	0.101
	RMA	0.096	0.059	0.098	0.060	0.137	0.060	0.120	0.059
	HIM	0.117	0.080	0.120	0.081	0.186	0.080	0.157	0.080
	Ours (w/ HIM)	0.091	0.061	0.093	0.061	0.139	0.061	0.117	0.061
	Ours	0.086	0.058	0.087	0.059	0.115	0.059	0.099	0.058



(a) Varying short-history length.



(b) Varying long-history length.

图4: 关于不同历史长度的反馈结构的消融研究。

比较我们使用预测模型的方法与全动力学模型, 以评估模型简化的效果。所有实验都涉及四旋翼无人机在三个随机种子下跟踪之字形轨迹。

1) *History structure*: 我们评估结合历史记录反馈结构。在短历史实验中, 我们将最近的输入/输出对, 包括预测误差, 串联成一个单一的输入向量。在长历史实验中, 我们保留最新的状态-动作对, 同时使用 TCN 模型[12], [14]对额外的历史进行编码, 并将其潜在输出与策略输入串联。

如图4所示, 使用更长的历史——无论是短期还是长期——并不会比即时反馈 (即最新的一步历史) 提高性能。虽然这与[14]有所不同, 但与 $\mathcal{L}_1$ 自适应控制[21]和鲁棒控制方法[2], [3]一致, 其中即时误差反馈足够。

表 IV: 内部预测模型的噪声耐受性评估, 在四旋翼跟踪之字形轨迹时进行测试。

	Noise σ_n	Train		Eval	
		static	disturbed	static	disturbed
	0.	0.057	0.064 (+12.28%)	0.101	0.119 (+17.82%)
input noise	0.02	0.063	0.069 (+9.52%)	0.092	0.117 (+27.17%)
	0.04	0.070	0.094 (+34.28%)	0.100	0.154 (+54.00%)
	0.08	0.071	0.121 (+70.42%)	0.134	0.202 (+50.74%)
	0.16	0.082	0.135 (+64.63%)	0.152	0.231 (+51.97%)
output noise	0.02	0.063	0.071 (+12.69%)	0.076	0.132 (+73.68%)
	0.04	0.073	0.097 (+32.87%)	0.102	0.161 (+57.84%)
	0.08	0.074	0.121 (+63.51%)	0.118	0.179 (+51.69%)
	0.16	0.076	0.130 (+71.05%)	0.138	0.222 (+60.86%)

用于提高精度和鲁棒性。这种更简单的结构也降低了实现的复杂性, 并减轻了实际机器人中的累积误差。

2) *Noise tolerance*: 为了评估内部预测模型精度对控制性能的影响, 我们在模型的状态输入或预测中引入不同程度的噪声。具体而言, 我们对状态向量 (包括位置、速度、四元数和角速度) 引入不同方差的零均值高斯噪声。对于四元数, 我们将采样的高斯噪声视为增量欧拉角, 将其转换为增量四元数, 然后应用于旋转原始姿态。我们使用一系列带噪的预测模型训练和评估多种策略, 以进行四旋翼在之字形轨迹上的跟踪, 结果如表 IV 所示。

结果表明, 我们的方法在轻度噪声下保持了有竞争力的性能, 无论是应用于模型输入还是输出。然而, 随着噪声水平的提高, 预测模型的准确性变得越来越关键, 尤其是在存在外部干扰或分布外系统动力学的场景中。在不同的训练和评估范围内, 随着噪声加剧, 性能下降, 并且在外力干扰下也观察到类似的趋势。这一观察突显了我们的显式预测模型相较于其他基于学习的方法的一个优势: 它提供了稳定的基线估计, 尽管是近似的。尽管建模噪声可能影响反馈的可靠性, 但刚体变换遵循基本运动原理, 并能在机器人系统中实现泛化。虽然该模型进行了简化。

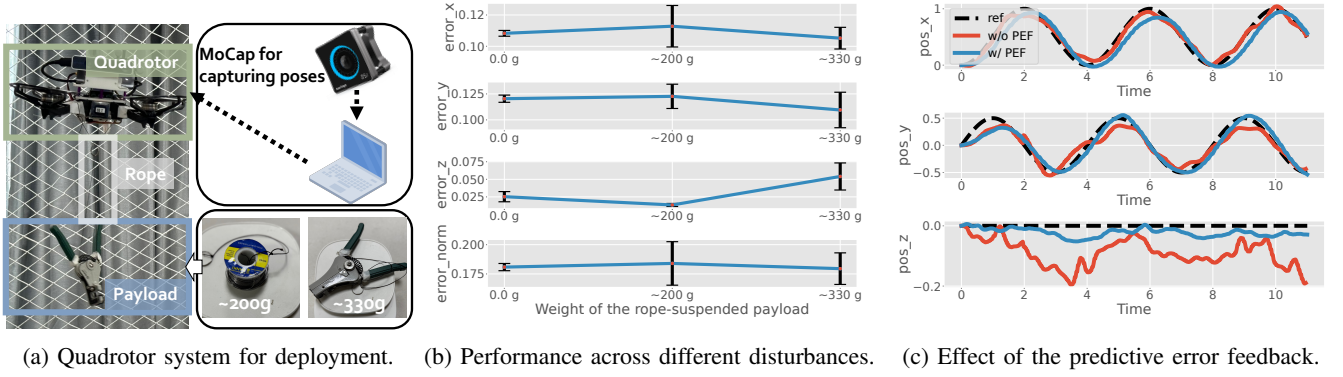


Fig. 5: Real-world experiments on a quadrotor with two different rope-suspending payloads.

TABLE V: Evaluation of the internal predictive model’s complexity, tested on quadrotors tracking zigzag trajectories.

Model	Train		Eval	
	static	disturbed	static	disturbed
Simplified	0.057	0.064 (+12.28%)	0.101	0.119 (+17.82%)
Full	0.061	0.067 (+9.84%)	0.095	0.105 (+10.53%)

nonlinear complexities, the RL part in our framework compensates effectively within an informative closed-loop system.

3) *Model simplification*: We investigate the impact of predictive model simplification on performance. A quadrotor serves as our testbed due to its well-studied dynamics model [40]. We compare our approach with a variant that uses the full-dynamics model for tracking zigzag trajectories. As shown in Tab. V, the results indicate that despite the simplification, performance degradation is marginal. Notably, this simplification improves generalization across different morphologies, making it applicable to complex systems that are difficult to model.

E. Real-world Deployment

To further validate the effectiveness of our framework, we conducted a real-world experiment on a quadrotor, deploying our method to assess its robustness under various disturbances. The quadrotor, equipped with an Nvidia Jetson Xavier NX [42] and a PX4 flight control unit (FCU) [43], interfaces with a motion capture system for real-time pose tracking. The whole neural control framework runs on the embedded computer. The control policy generates CTBR commands, transmitted to the FCU via MAVROS [44] and converted to motor velocities. The quadrotor weighs approximately 1 kg, and we evaluated its tracking performance across five trials under three disturbance conditions: no disturbance, a 200g payload, and a 330g payload, each attached via a rope to increase instability and nonlinearity. The deployment system setup is shown in Fig. 5a.

We present the tracking performance for each axis, along with the normalized error across all axes, in Fig. 5b. The lines represent the average tracking errors under different conditions, with black intervals indicating variance. Our results show that the framework maintains relatively stable

performance across various payloads. To illustrate the impact of predictive error feedback, we include a detailed example trajectory for the 330g payload as an ablation study in a real-world scenario (Fig. 5c). The z-axis trajectory demonstrates that, without predictive error feedback, the control policy struggles to maintain a stable flight height due to the unexpected heavy payload. In contrast, our method, which integrates predictive error feedback, achieves robust tracking despite the downward force from the additional payload.

V. CONCLUSION

In this paper, we propose Neural Internal Model Control (NeuralIMC), a framework that bridges model-based control with RL-based neural control. By leveraging body dynamics and a white-box rigid-body transformation, we avoid high-dimensional nonlinearity, using a simplified predictive model in a closed-loop feedback structure. Model-free RL enables the policy to react to predictive errors. NeuralIMC outperforms state-of-the-art controllers on quadrotors and quadrupedal robots. Real-world tests on a heavily loaded quadrotor demonstrate robustness to disturbances and dynamic variations, highlighting its potential for interpretable and adaptive learning-based control. For the future work, we’d like to conduct more expanded real-world experiments includes testing on diverse robots and improving disturbance training [45].

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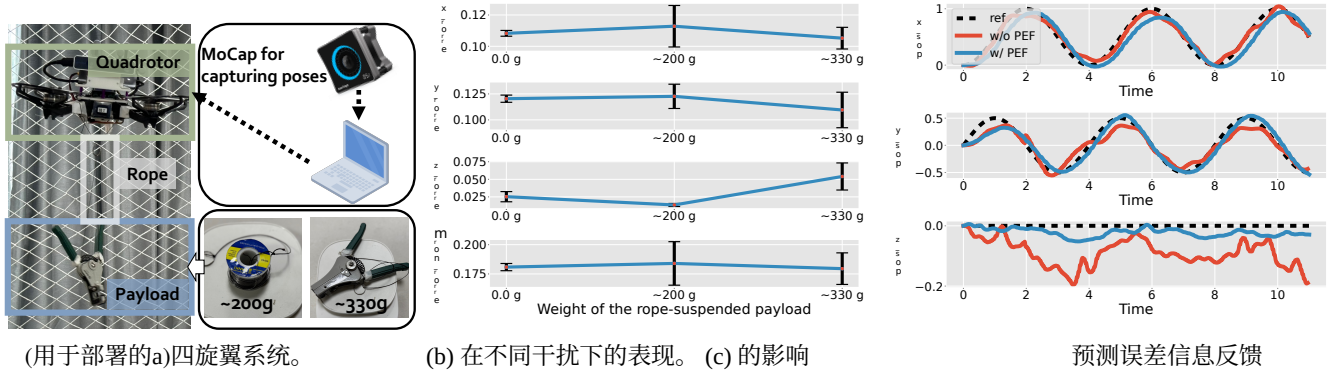


图5：在一台四旋翼上进行的涉及两种不同绳索悬挂载荷的实际实验。

表 V：内部预测模型复杂性的评估，在四旋翼无人机跟踪之字形轨迹时进行测试。

Model	Train		Eval	
	static	disturbed	static	disturbed
Simplified	0.057	0.064 (+12.28%)	0.101	0.119 (+17.82%)
Full	0.061	0.067 (+9.84%)	0.095	0.105 (+10.53%)

非线性复杂性，我们框架中的强化学习部分在一个信息闭环系统中能够有效补偿。

3) *Model simplification*: 我们研究了预测模型简化对性能的影响。四旋翼作为我们的测试平台，因为其动力学模型研究充分[40]。我们将我们的方法与使用完整动力学模型跟踪之字形轨迹的变体进行了比较。如表V所示，结果表明，尽管进行了简化，性能下降是有限的。值得注意的是，这种简化提高了在不同形态间的泛化能力，使其适用于难以建模的复杂系统。

E. Real-world Deployment

为了进一步验证我们框架的有效性，我们在四旋翼上进行了真实环境实验，部署我们的方法以评估其在各种扰动下的鲁棒性。该四旋翼配备了 Nvidia Jetson Xavier NX [42] 和 PX4 飞控单元 (FCU) [43]，并通过运动捕捉系统进行实时位姿跟踪。整个神经控制框架运行在嵌入式计算机上。控制策略生成 CTBR 指令，通过 MAVROS [44] 传输到 FCU 并转换为电机速度。四旋翼重量约为 1 公斤，我们在三种扰动条件下进行了五次试验来评估其跟踪性能：无扰动、200 克负载和 330 克负载，每个负载通过绳索连接以增加不稳定性及非线性。部署系统的设置如图 5a 所示。

我们在图 5b 中展示了每个轴的跟踪性能，以及所有轴的归一化误差。线条表示不同条件下的平均跟踪误差，黑色区间表示方差。我们的结果显示，该框架保持了相对稳定的

在各种有效载荷下的性能。为了说明预测误差反馈的影响，我们在真实场景中为 330 克有效载荷提供了一个详细的示例轨迹，作为消融研究（图 5c）。z 轴轨迹表明，如果没有预测误差反馈，控制策略由于意外的重载而难以维持稳定的飞行高度。相比之下，我们的方法融合了预测误差反馈，尽管额外有效载荷产生向下的力，但仍能实现稳定的跟踪。

五、结论

在本文中，我们提出了神经内部模型控制 (NeuralIMC)，一个将基于模型的控制与基于强化学习的神经控制相结合的框架。通过利用身体动力学和白盒刚体变换，我们避免了高维非线性，使用简化的预测模型在闭环反馈结构中。无模型强化学习使策略能够对预测误差作出反应。NeuralIMC 在四旋翼和四足机器人上的表现优于最先进的控制器。在负载较大的四旋翼上的实际测试表明其对干扰和动态变化具有鲁棒性，突显了其在可解释且自适应的基于学习的控制中的潜力。对于未来的工作，我们希望进行更多扩展的实际实验，包括在多样化机器人上的测试以及改进干扰训练 [45]。

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